

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1 set up workspace and data
```

```
library(tidyverse); library(here); library(cowplot)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    4.0.0      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
## here() starts at /home/guest/EDE_Fall2025
##
##
## Attaching package: 'cowplot'
##
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
here(); getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
lake.nutrient.data <- read.csv(
  file=here('./Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv'),
  stringsAsFactors = TRUE)
lake.chemphys.data <- read.csv(
  file=here('./Data/Processed_KEY/NTL-LTER_Lake_ChemistryPhysics_PeterPaul_Processed.csv'),
  stringsAsFactors = TRUE)
litter.data <- read.csv(
  file=here('./Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv'),
  stringsAsFactors = TRUE)

#2 dates from factors
lake.nutrient.data$sampldate <- ymd(lake.nutrient.data$sampldate)
lake.chemphys.data$sampldate <- ymd(lake.chemphys.data$sampldate)
litter.data$collectDate <- ymd(litter.data$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3 make a theme
mytheme <- theme_gray() +
  theme(legend.position = "bottom",
        legend.direction = "horizontal",
        plot.title = element_text(hjust = 0.5))

theme_set(mytheme)
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

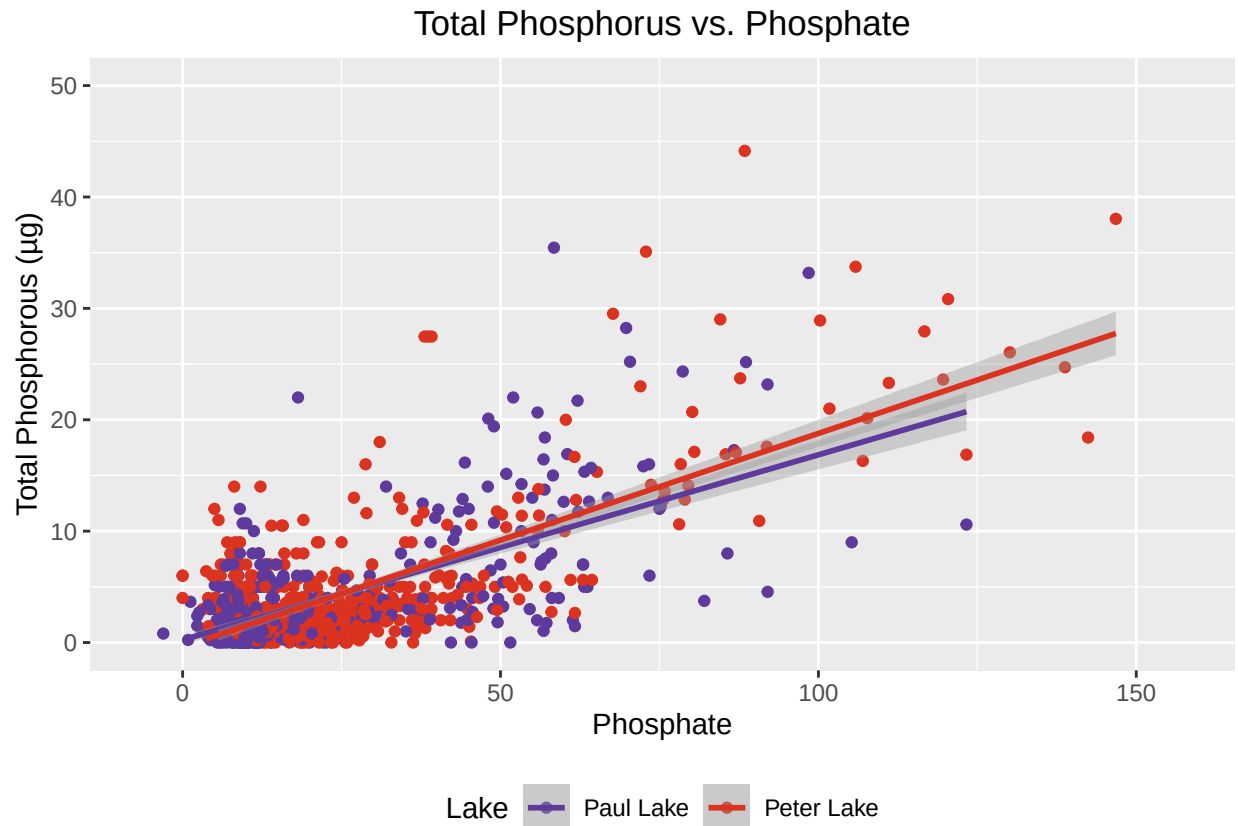
```
#4
ggplot(lake.nutrient.data) +
  aes(x = tp_ug, y = po4, color = lakename) +
  scale_color_manual(values = c("#5D3A9B", "#DC3220")) + #colorblind accessible
  ylim(c(0,50)) +
  geom_point() +
  geom_smooth(method=lm) +
  labs(
    title = "Total Phosphorus vs. Phosphate",
    color = "Lake",
    x = "Phosphate",
    y = "Total Phosphorous (µg)")

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 21947 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 21947 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
temp.by.month <-
  ggplot(lake.nutrient.data) +
    aes(x = as.factor(month), y = temperature_C, color = lakename) +
    scale_color_manual(values = c("#5D3A9B", "#DC3220")) +
    geom_boxplot() +
    labs(
      x = element_blank(),
      y = "Temp (°C)",
      title = "Temperature, Total Phosphorous, and Total Nitrogen by Month") +
    theme(legend.position = "none")

TP.by.month <-
  ggplot(lake.nutrient.data) +
    aes(x = as.factor(month), y = tp_ug, color = lakename) +
    scale_color_manual(values = c("#5D3A9B", "#DC3220")) +
```

```
geom_boxplot() +
labs(
  x = element_blank(),
  y = "TP (µg)" +
ylim(c(0,100))
theme(legend.position = "none")
```

```
## <theme> List of 1
## $ legend.position: chr "none"
## @ complete: logi FALSE
## @ validate: logi TRUE
```

```
TN.by.month <-
ggplot(lake.nutrient.data) +
aes(x = as.factor(month), y = tn_ug, color = lakename) +
scale_color_manual(values = c("#5D3A9B", "#DC3220")) +
geom_boxplot() +
labs(
  x = "Month Number",
  y = "TN (µg)",
  color = "Lake" +
ylim(c(0,1250))

plot_grid( # using 'null' plots with negative heights to reduce empty space
  temp.by.month, NULL, TP.by.month, NULL, TN.by.month,
  nrow = 5, align = "h", rel_heights = c(1,-0.3,1.25,-0.3,1.25))
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

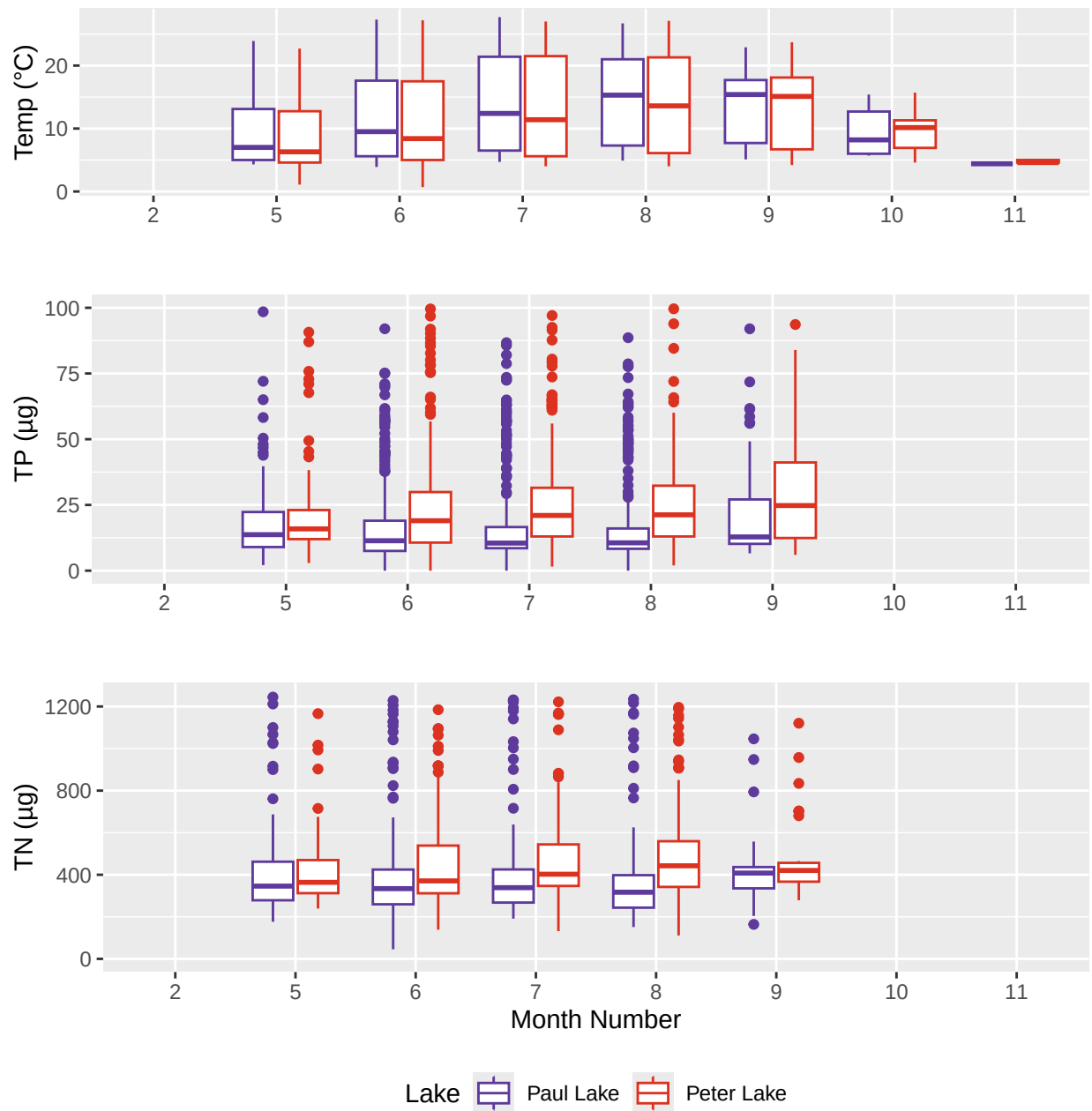
```
## Warning: 'label' cannot be a <ggplot2::element_blank> object.
```

```
## Warning: Removed 20799 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: 'label' cannot be a <ggplot2::element_blank> object.
```

```
## Warning: Removed 21676 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

Temperature, Total Phosphorous, and Total Nitrogen by Month



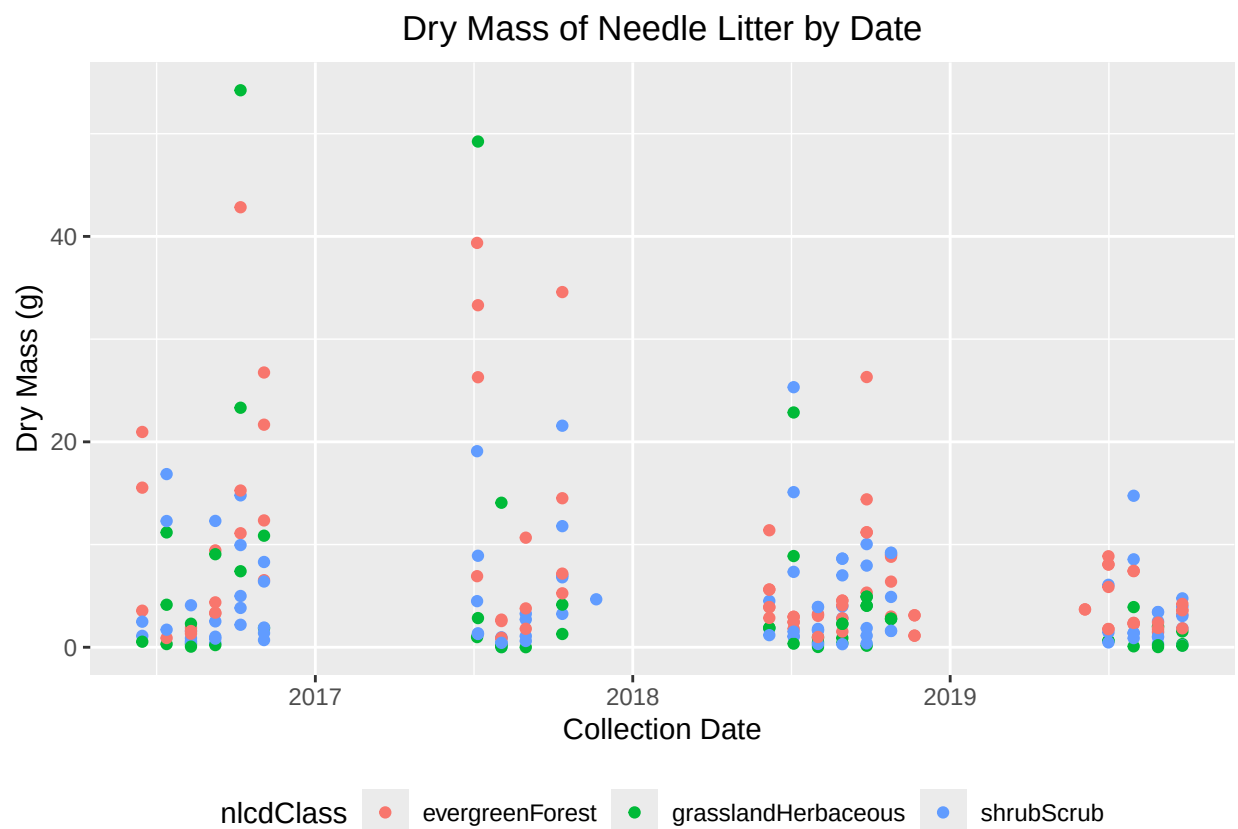
Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: As would be expected, temperature appears to rise during the summer months in both lakes. Summer temperature measurements also seem to have the greatest variability. Overall, Peter Lake shows slightly higher readings for both nutrients. Across the seasons, there is not much change in nutrient levels in Paul Lake until increased levels of TP and TN in September. TP and TN levels in Peter Lake seem to climb steadily as summer progresses.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)

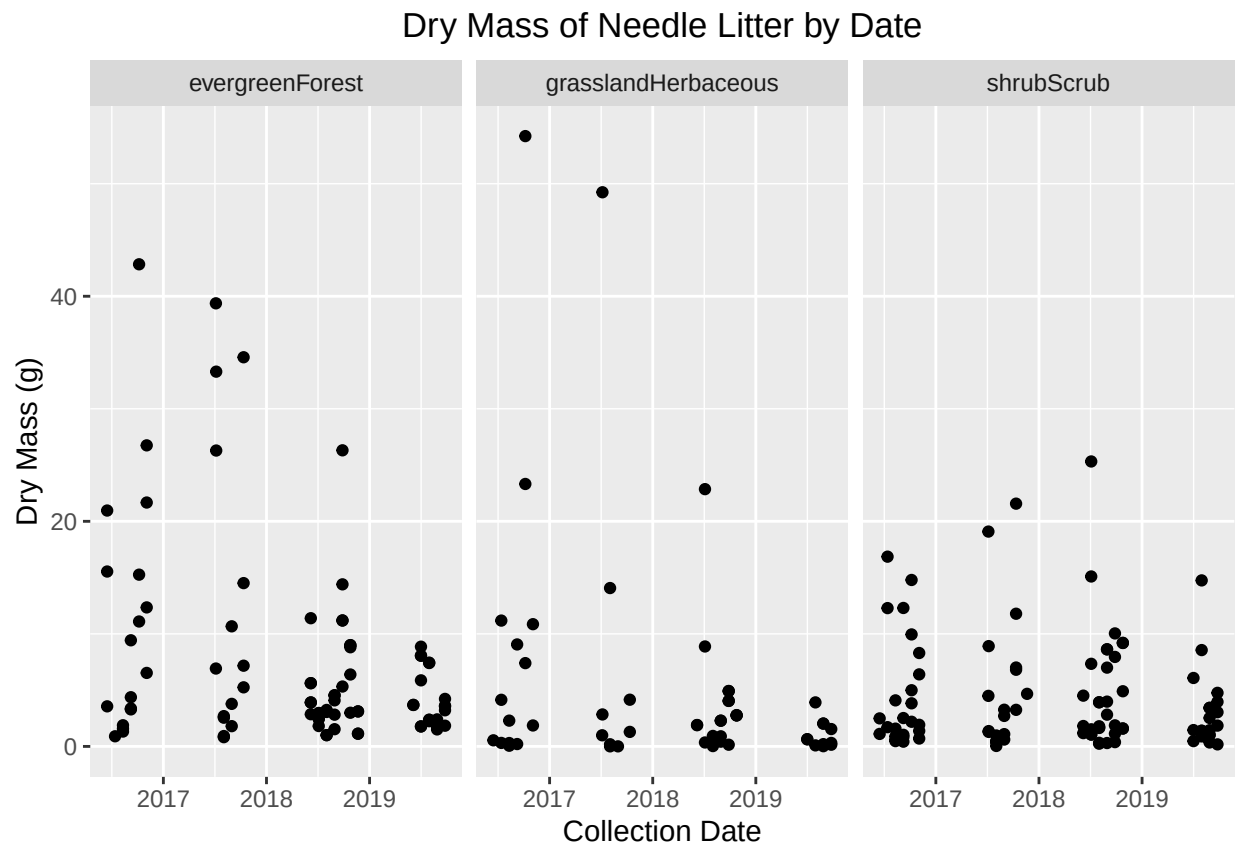
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
litter.data %>%
  filter(functionalGroup == "Needles") %>%
  ggplot() +
  aes(x = collectDate, y = dryMass, color = nlcdClass) +
  geom_point() +
  labs(
    title = "Dry Mass of Needle Litter by Date",
    x = "Collection Date",
    y = "Dry Mass (g)"
  )
)
```



```
#7
litter.data %>%
  filter(functionalGroup == "Needles") %>%
  ggplot() +
  aes(x = collectDate, y = dryMass) +
  geom_point() +
  facet_wrap(vars(nlcdClass), ncol = 3) +
  labs(
    title = "Dry Mass of Needle Litter by Date",
    x = "Collection Date",
  )
```

```
y = "Dry Mass (g)"
)
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: Plot 7 is more effective. In plot 6, there are too many colors and the points are crowded too densely to draw any meaningful interpretations. In plot 7, the eye can focus on one cluster of related data at a time for better understand of its distribution, then compare clusters across sub-plots.